

# DESIGN GUIDELINES

## AI TRIAGE SUPPORT SYSTEM CO-DESIGN CANVAS

What things are important to consider in the robot's design?

### Advantage guidelines

What advantages should the system provide?

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- Support, not replace, nurse clinical judgement.
- Improve consistency of ESI recommendations across different staff and shifts.
- Reduce cognitive workload during busy Emergency Department periods.
- Present AI recommendations with confidence scores and contributing factors.
- Allow rapid confirmation, override and reassessment.
- Improve communication between waiting patients and nurses through the patient portal.
- Maintain complete audit trails for governance and quality improvement.
- Integrate naturally into existing triage workflows without delaying patient care.

### Ethical guidelines

What ethical considerations should guide the design?

- AI recommendations must remain advisory only.
- Final triage decisions remain the responsibility of the nurse.
- Patient privacy must be protected through secure authentication and encryption.
- All significant actions, overrides and reassessments must be recorded.
- The system should minimise algorithmic bias through continuous monitoring and validation.
- Patients should understand that AI supports, but does not replace, clinical staff.
- Accessible alternatives should be available for patients unable to use smartphones.
- Manual workflows must remain available if the system becomes unavailable.

## SYSTEM DIMENSIONS

### Environment guidelines

How should the system fit its environment?

- Designed for busy, high-pressure Emergency Department workflows.
- Support use during long shifts, interruptions and periods of fatigue.
- Require minimal additional data entry.
- Remain usable under noisy, crowded clinical conditions.
- Operate securely on hospital networks.
- Continue supporting manual workflows during outages.
- Allow future integration with the Mercer Electronic Health Record and approved clinical devices.
- Support patient communication while they wait for reassessment.

### Form guidelines

How should the system look? | qualities?

- Clean, uncluttered interface prioritising essential clinical information.
- High-contrast colours with large, readable text.
- Colour always paired with text or icons.
- Clearly distinguish nurse decisions from AI recommendations.
- Consistent layout across all screens.
- Mobile-friendly patient portal using simple, non-technical language.
- Accessible controls with minimal navigation steps.
- Professional clinical appearance suitable for hospital use.

### Interaction guidelines

How should users interact with it?

- Nurses remain in control of every clinical decision.
- AI recommendations require review before acceptance.
- Patient symptom updates should notify nurses without automatically changing priority.
- Allow rapid confirmation, override and reassessment.
- Minimise the number of clicks required during busy shifts.
- Support both desktop and mobile interactions where appropriate.
- Provide clear feedback after every user action.
- Prioritise task completion over unnecessary interaction.

### Behaviour guidelines

How should the system behave? | r?

- Provide recommendations only after sufficient clinical information is available.
- Display confidence scores and contributing clinical factors.
- Highlight low-confidence predictions for additional review.
- Prioritise clinically significant alerts while reducing unnecessary notifications.
- Escalate patient deterioration reports to the nurse dashboard.
- Record recommendations, overrides and reassessments automatically.
- Continue operating safely during partial failures by supporting manual workflows.
- Behave consistently and predictably across all users and clinical shifts.